

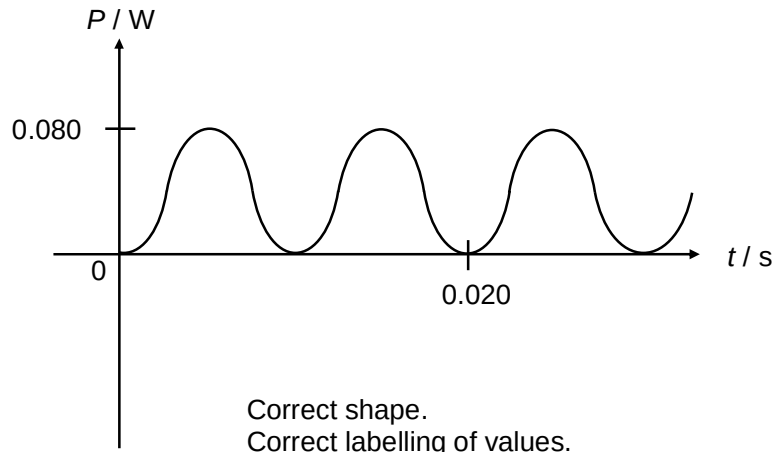
6 (a) $V_{r.m.s.} = \sqrt{\frac{2^2(0.002) + 1^2(0.002)}{0.01}}$ M1
 $= 1.0 \text{ V}$ A1

Steady voltage of 1 V will produce the same heating effect as $V_{r.m.s.}$ of 1 V.

- (b) Transmission of electrical energy at high voltage means that the current is low according to $P = IV$. B1
 Power loss through joule heating (I^2R) is hence lowered as less electrical energy is dissipated as heat in the cables of resistance R . B1

(c) (i) $\frac{V_s}{V_p} = \frac{N_s}{N_p}$
 $V_s = 71 \times 6.5 \times 10^{-3} = 0.46 \text{ V}$ A1

(ii)



- (iii) 1. In the forward biased direction, the diode has no resistance. Current flows downwards through resistor R . A1
 In the reverse biased direction, diode has infinite resistance. There is no current flowing through resistor R . A1
 2. In the forward biased direction, there is a half-wave sinusoidal voltage output across resistor R , having the same frequency as that of the input voltage. In the reverse biased direction, there is no voltage output across resistor R . A1

7 (a) Electromotive force is the work done in transforming non-electrical energy into electrical